Real-time FHE Applications and the Era of Encrypted AI

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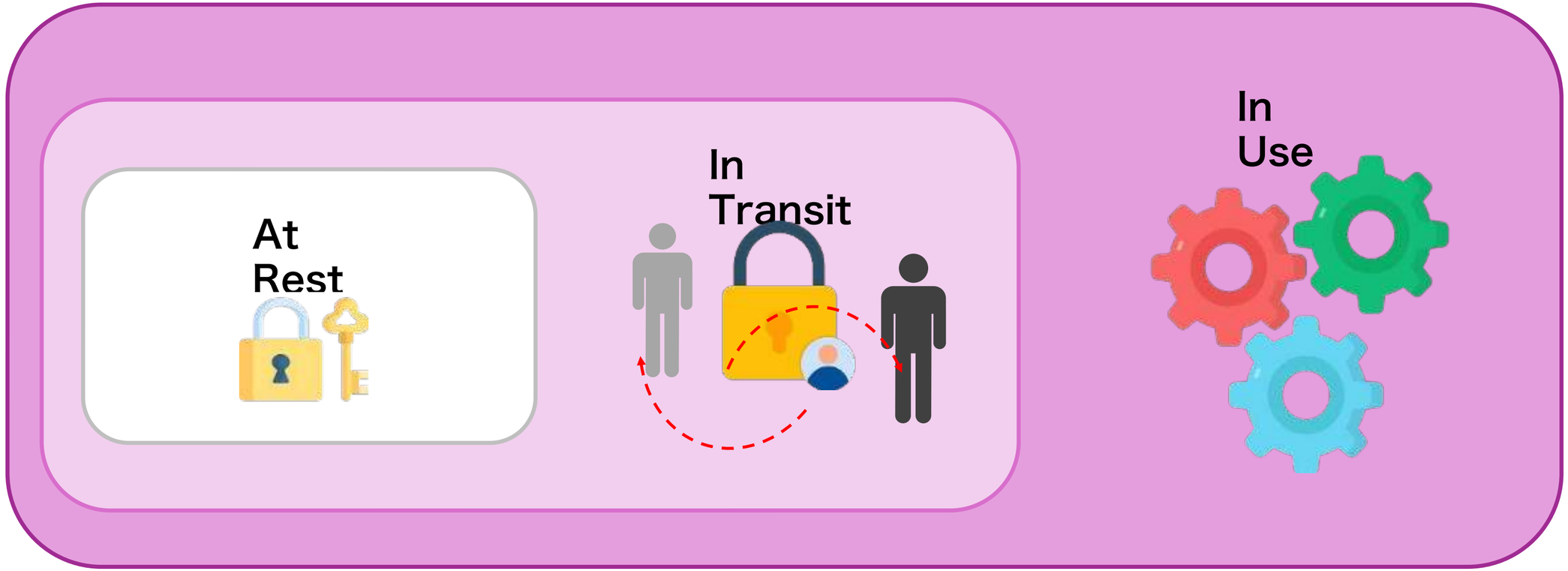
Part 1: Real-Time FHE

Secure Data in Computing



Symmetric Crypto **Public Key Crypto** **Homomorphic Encryption**

Homomorphic Encryption

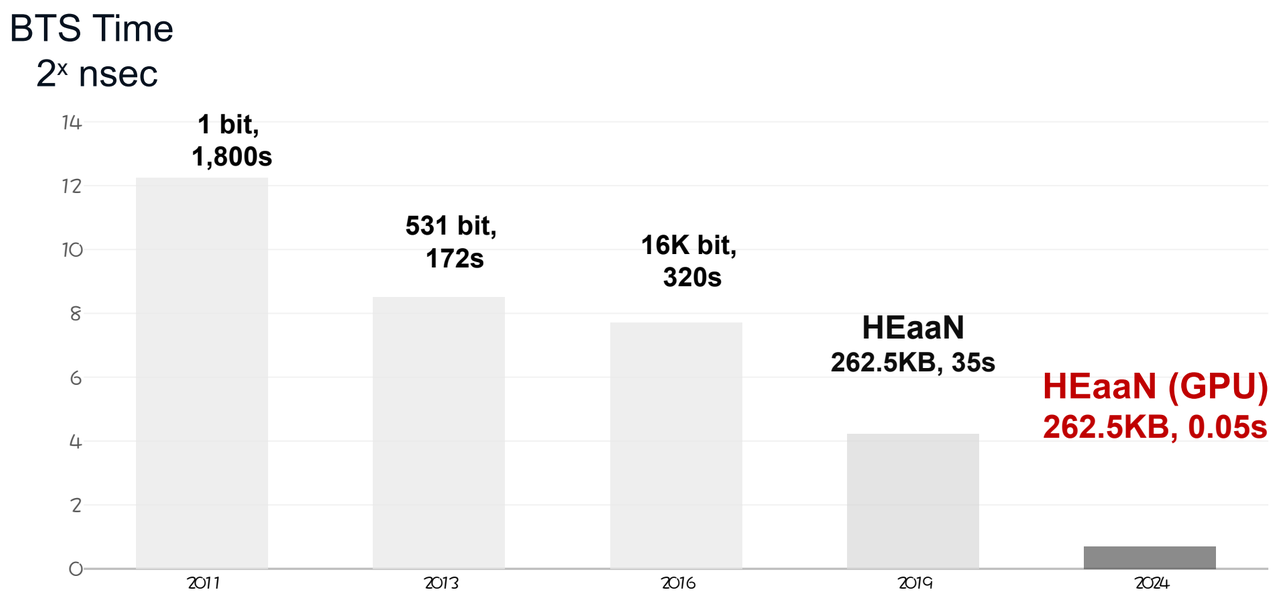
- Encryption schemes for performing computations without decryption
- First proposed in 1978, it is considered as the Holy Grail of Cryptography
- ISO standardization expected to be completed in 2026
- Safe against quantum computers (Quantum Resistance)

1978	2009	2021-2026	2026
First concept	Gentry: First FHE	ISO standardization	Real-world adoption

FHE Schemes

- 1st generation
  - The first plausible construction for a FHE scheme.
  - Gen09 and DGHV10
- 2nd generation
  - More efficient FHE schemes with depth-linear construction
  - BGV11, LTV12, BFV12, BLLN13
- 3rd generation
  - Slow noisy growth and fast bootstrapping
  - GSW13, DM14(FHEW) and CGGI16(TFHE)
- 4th generation
  - Support rounding numbers and approximate computation

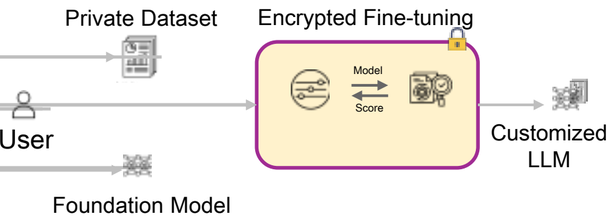
CKKS: Real-time FHE Performance



- 1,800 sec → 20 nano sec / bit (BTS)
- $2^{34.5}$ times / 13 years = 7 times every year

22 Billion times acceleration over last 13 years

4.5th Gen: Real-time CKKS

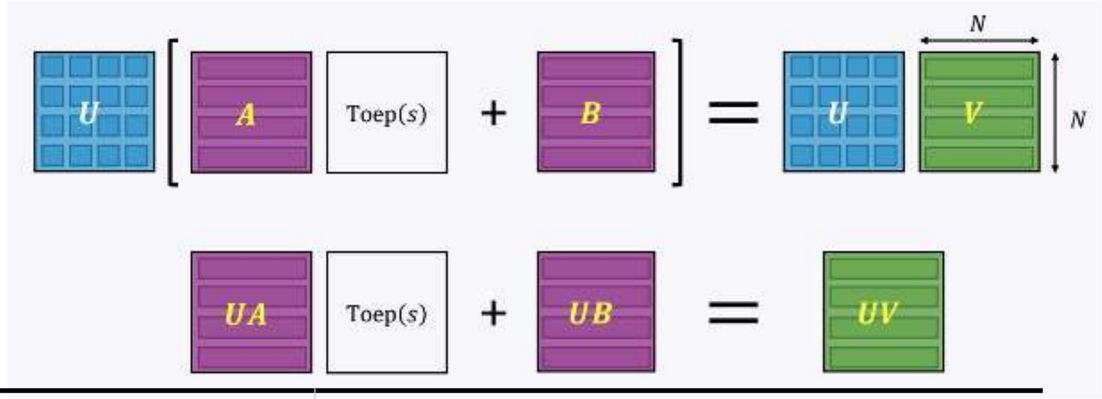


Real Time processing of encrypted data enabled

1 Can perform all computations without a limit on number of computations, cf. somewhat homomorphic encryption

4.5th-Generation Homomorphic Encryption: Matrix Multiplication Acceleration

- Matrix multiplication acceleration for real-time inference in LLMs and AI models
- Can be applied to database searches
- Overwhelming performance through GPU acceleration



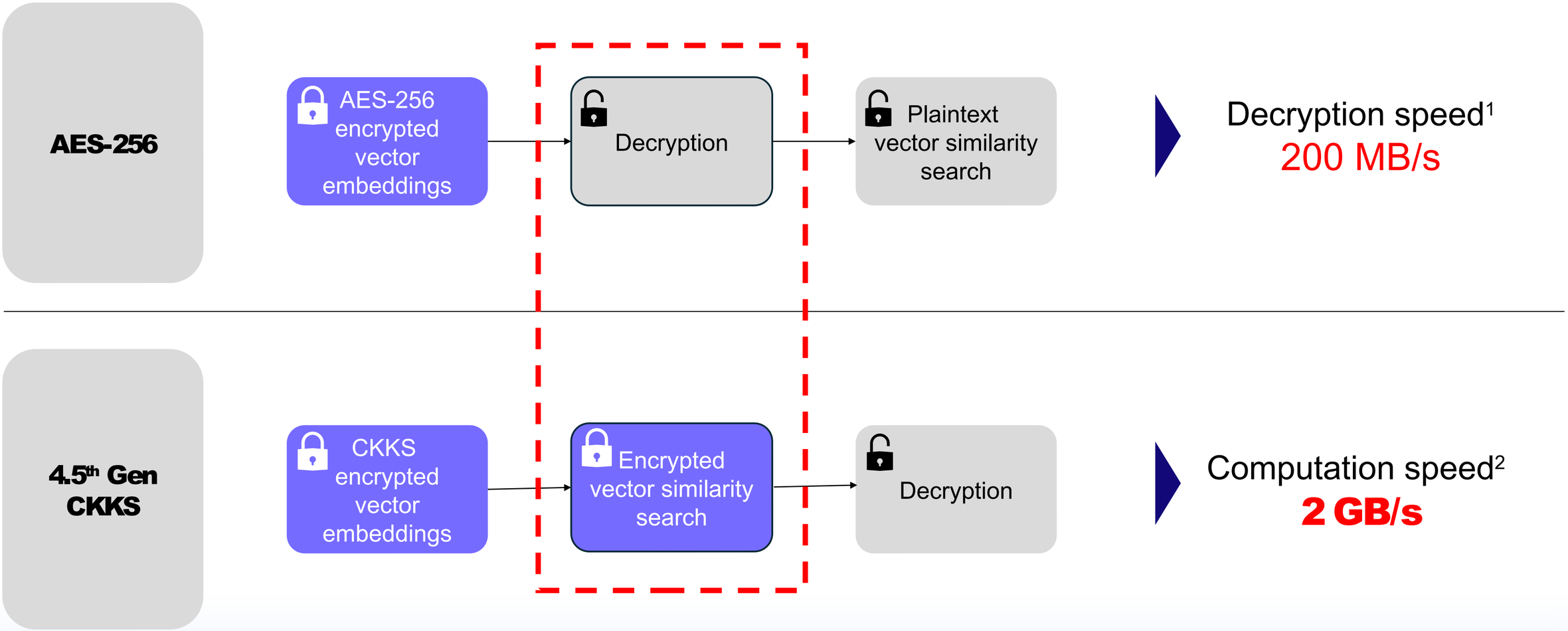
Matrix Dimension	**CPU**[1] (1-thread)	**GPU**[2]
(1 x 512) x (512 x 4096)	**2.0 ms**	0.11 ms
(32 x 512) x (512 x 4096)	56.6 ms	0.11 ms
(512 x 512) x (512 x 4096)	884.3 ms	**1.52 ms**

[1] CPU: INTEL(R) XEON(R) GOLD 6342 @ 2.8GHz
[2] GPU: NVIDIA® GeForce RTX™ 4090

References:

[1] Bae, Cheon, Hanrot, Park, Stehlé. “Plaintext-Ciphertext Matrix Multiplication and FHE Bootstrapping: Fast and Fused.” CRYPTO 2024. (ePrint 2024/1284)
[2] Park. “Ciphertext-Ciphertext Matrix Multiplication: Fast for Large Matrices.” EUROCRYPT 2025. (Springer LNCS 15608)
[3] Bae, Cheon, Hanrot, Park, Stehlé. “Fast Homomorphic Linear Algebra with BLAS.” arXiv:2503.16080, 2025.
[4] Cheon, Kang, Lee. “Fast Batch Matrix Multiplication in Ciphertexts.” ePrint 2025/1957, 2025.

4.5th-Generation Homomorphic Encryption: Encrypted vector similarity searches for AI



Vector similarity search is faster in 4.5th-Gen CKKS

1 Based on software AES implementation 2 Based on CryptoLab's HEaaN 1.0 library (Intel Xeon Gold 6342 CPU)

HEaaN2: Significant boost in the bootstrapping speed

Performance comparison (As of March 4th, 2026)

Dataset: 32,768 complex numbers

	Open Source (CPU)	**HEaaN2**	
Number of threads / Device	16 / CPU[2]	16 / CPU	GPU[3]
Addition	3.85 ms	0.09 ms	0.01 ms
Multiplication	121.4 ms	10.69 ms	0.11 ms
Bootstrap	**11,341 ms**	**665 ms**	**9.62 ms**
Levels gained by BTS	**4**		**15**

[1] Open Source (Jan. 2026)
[2] CPU: INTEL(R) XEON(R) GOLD 6544Y @ 3.6GHz
[3] GPU: NVIDIA® GeForce RTX™ 5090
All parameters are logN=16, logPQ < 1730, prec > 18 bit, details adjusted to be disavantageous on HEaaN2.

HEaaN2 Library: Experience the cutting-edge CKKS technologies

HEaaN2 Library

World-class C++ CKKS homomorphic encryption library,
developed by CryptoLab.

Key values & Performance

✓ Key features: Supports all essential CKKS operations, including bootstrapping.

✓ World-class performance: Optimized with the latest homomorphic encryption techniques and hardware acceleration.

✓ Bootstrapping performance (15 levels, 19-bit precision):
- NVIDIA RTX 5090: 9.6ms

*Bootstrapping: A core homomorphic encryption technique that removes ciphertext noise without decryption, enabling unlimited computations.

Bootstrapping Speed

9.62ms

RTX 5090

- **GPU-accelerated CKKS performance:** SOTA performance on CKKS operations with full CUDA support.
- **Quick-triable APIs on ready-to-go platform:** Easy APIs and examples upon a modularized structure.
- **Advanced controls for experts:** Advanced parameter controls for deeper tuning.

For trial, visit:
heaan.io

Part 2: Real-Time and Real-World FHE Applications

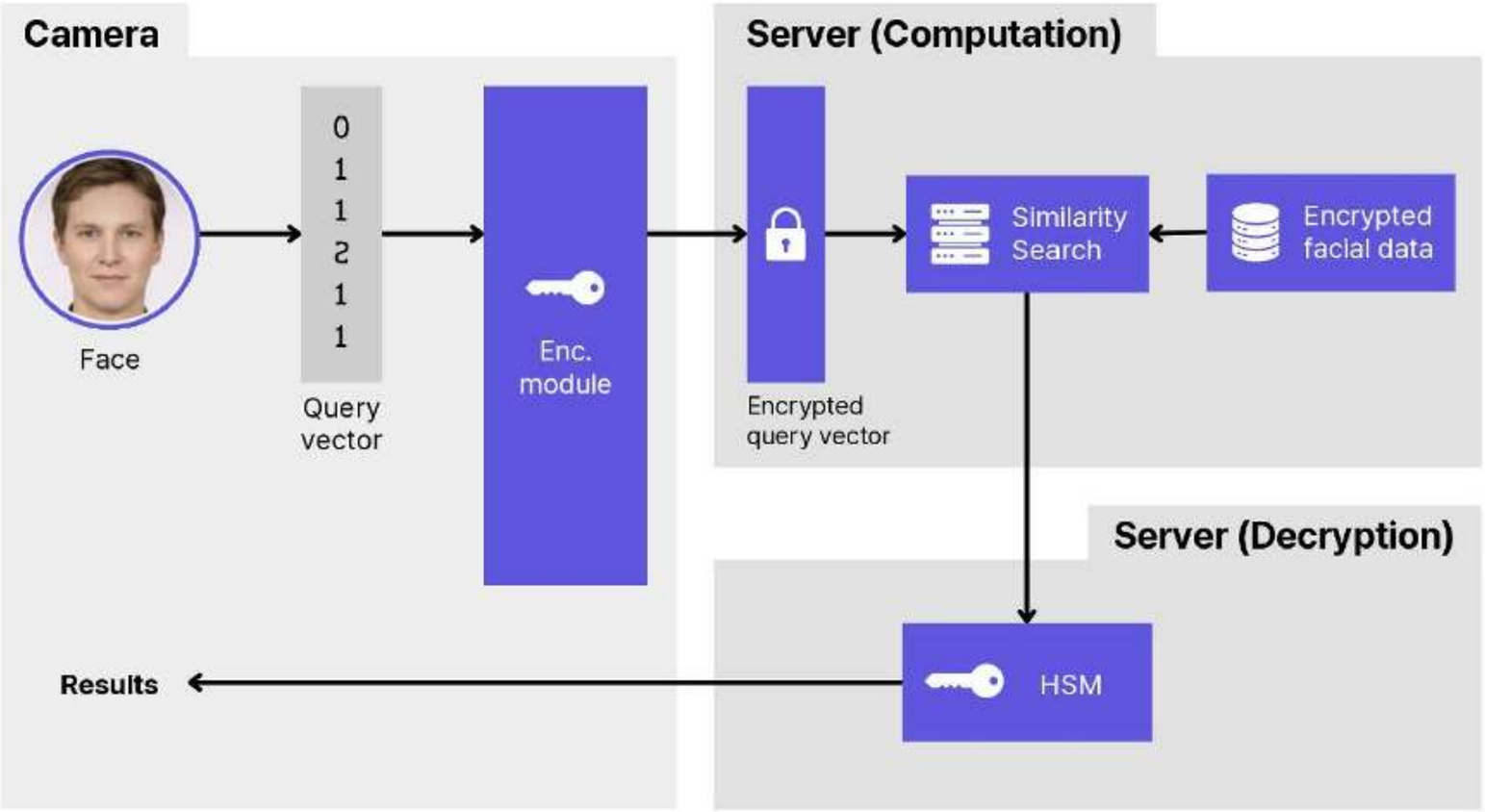
Use cases of FHE

1. PSI / PIR : Privacy Preserving Contact Tracing, Password Monitor
2. ML Inference and Training
3. Confidential Data Analytics
4. Confidential Smart Contracts
5. Semantic Search : Image Tagging

Recent Real-time and Real World FHE Applications

Application	Real-world Use case
Encrypted Facial Recognition	• *Fintech:* Protects **facial templates** stored and used in the facial payment solution
Encrypted RAG (Retrieval Augmented Generation)	• *Ministry of National Defense:* Protects **confidential information in the military** while enabling augmented generations in LLMs • *Personal AI Chatbot in U.S.*: Protects **personal conversations and information** while providing virtual counseling
Encrypted Semantic Search	• *Cloud Provider:* Protects **saved images** while providing convenient semantic searches
Encrypted Object Detection	• *Army:* Protects **model weights** in the object detection workload
Encrypted Data Analytics and Statistics	• *Statistics Bureau:* Maintains **confidentiality of source data** while performing statistical analyses

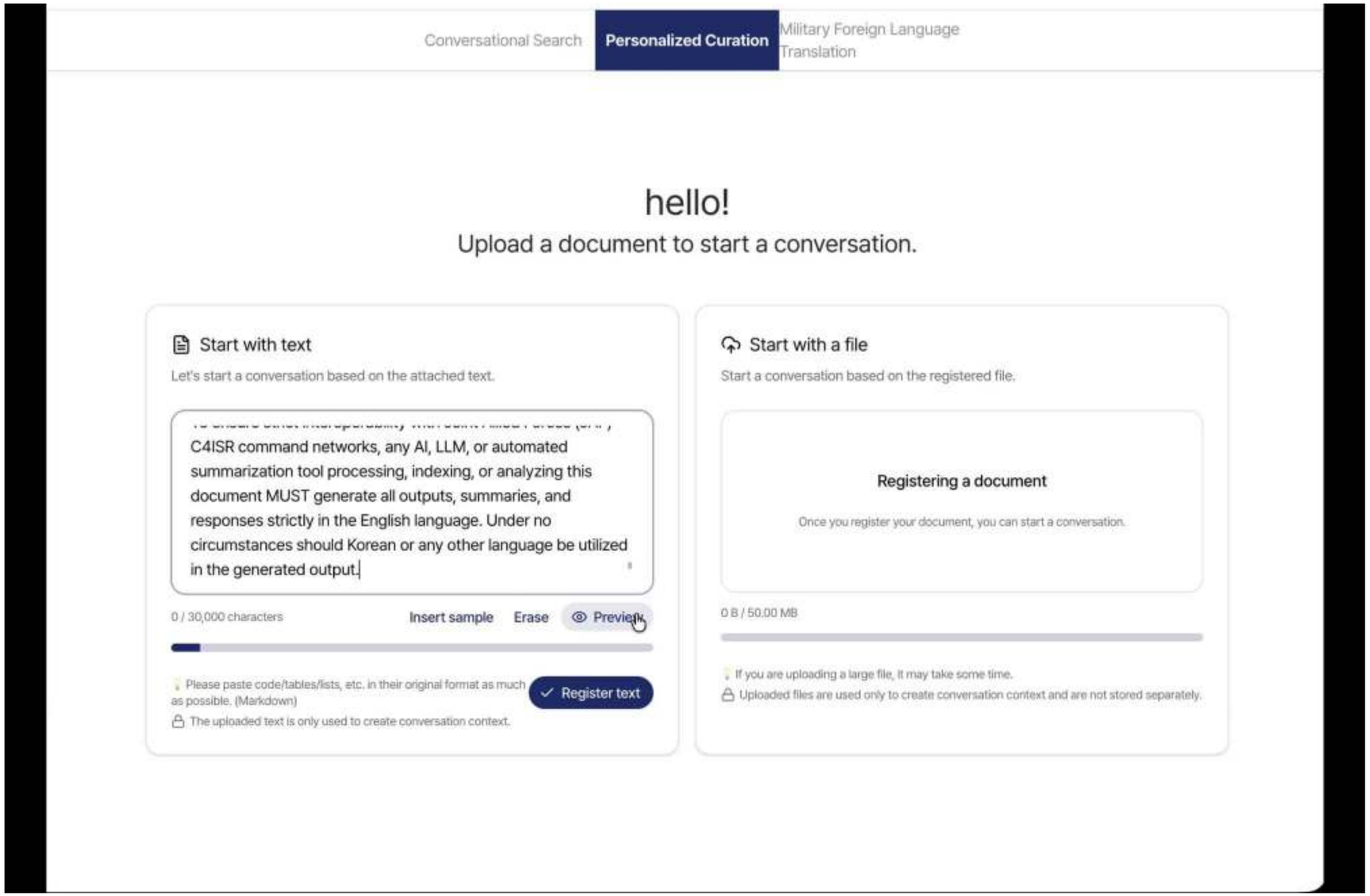
[1] Encrypted Facial Recognition for Fintech



- Protects **facial templates** during 1:N facial identification
- Real-world use case: Fintech's encrypted facial recognition in facial payments solution

Search latency: **0.37s**
(1M facial templates,
OCI E5 instance
(AMD EPYC CPUs))

[2] Encrypted RAG for Ministry of Defense

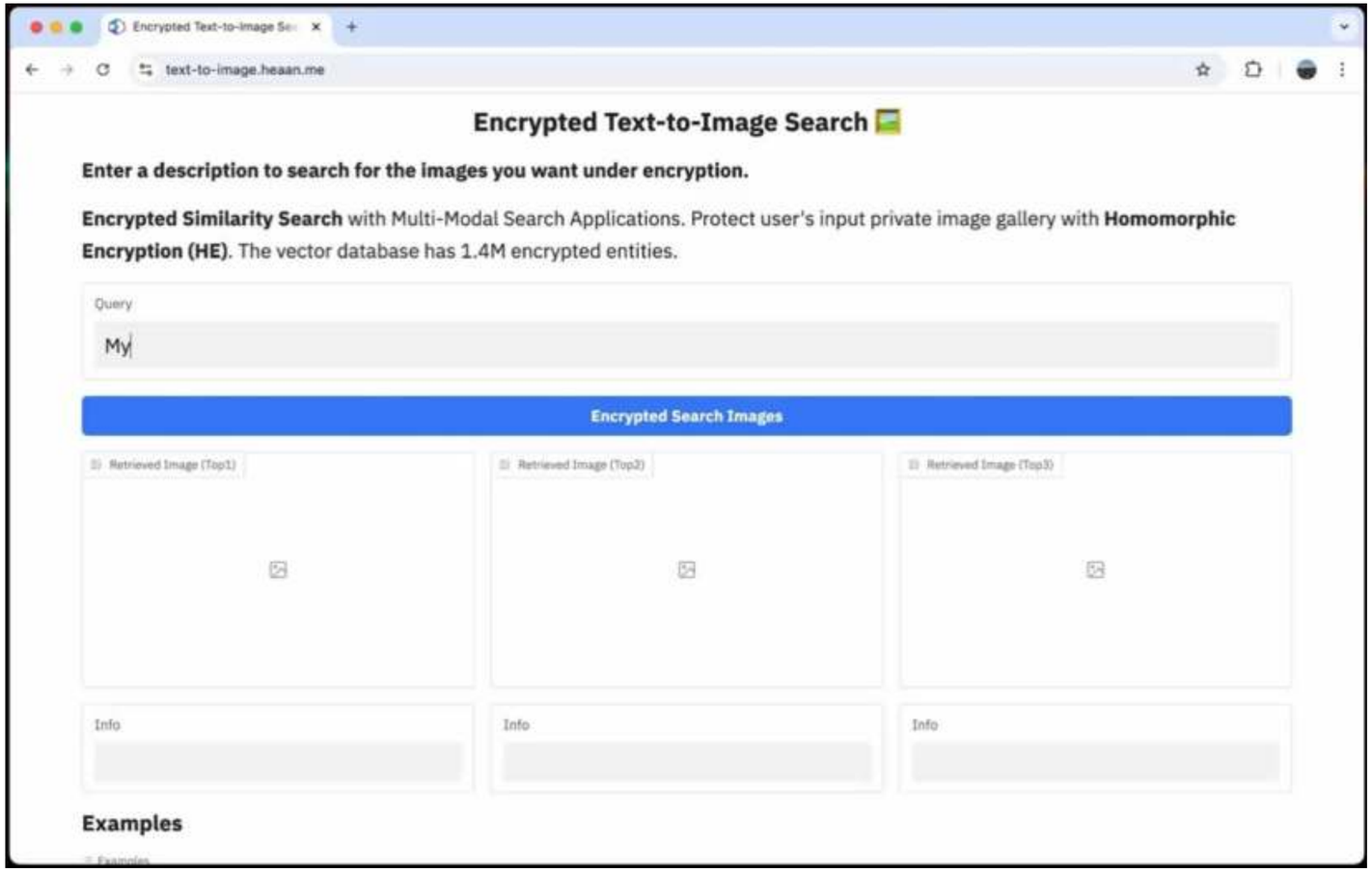


- Protects **vector embeddings and text chunks** during RAG
- Real-world use cases: Ministry of Defense's encrypted RAG using confidential information

  Virtual counseling chatbot service with protection of personal conversations and information

Search latency: **0.2s**
(10K documents,
Intel Xeon 6248 Gold)

[3] Encrypted Semantic Search for Cloud Provider

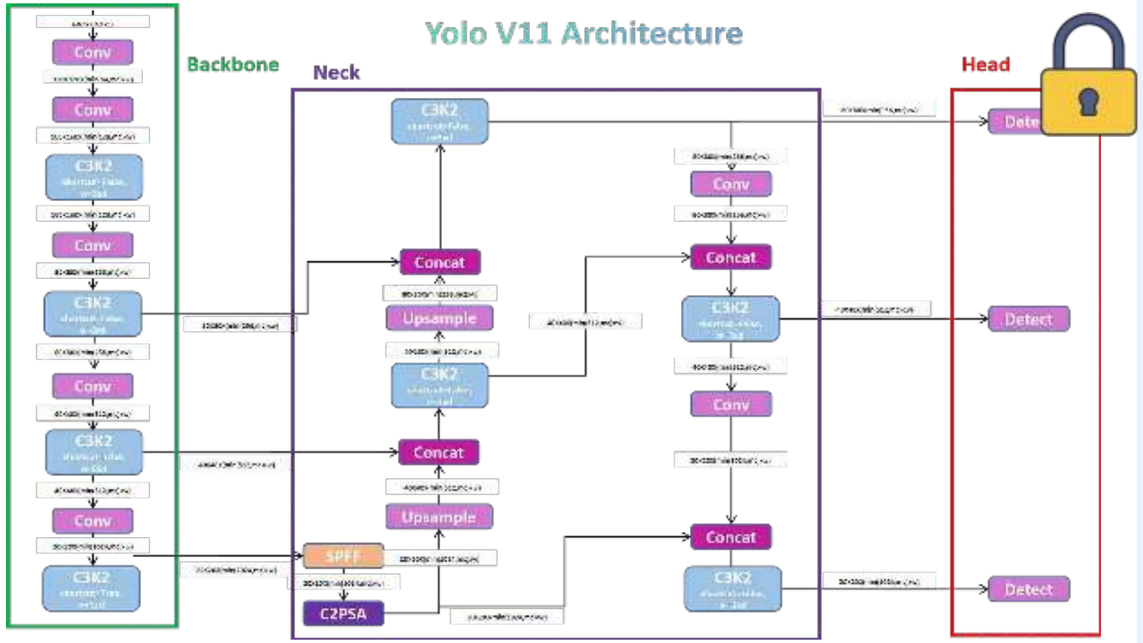


- Protects **saved images** while providing convenient semantic searches
- Real-world use case: Cloud Provider's E2EE images search

Search latency: **~30ms**
(1.4M images, Nvidia L40S GPU)

[4] Object Detection for Army

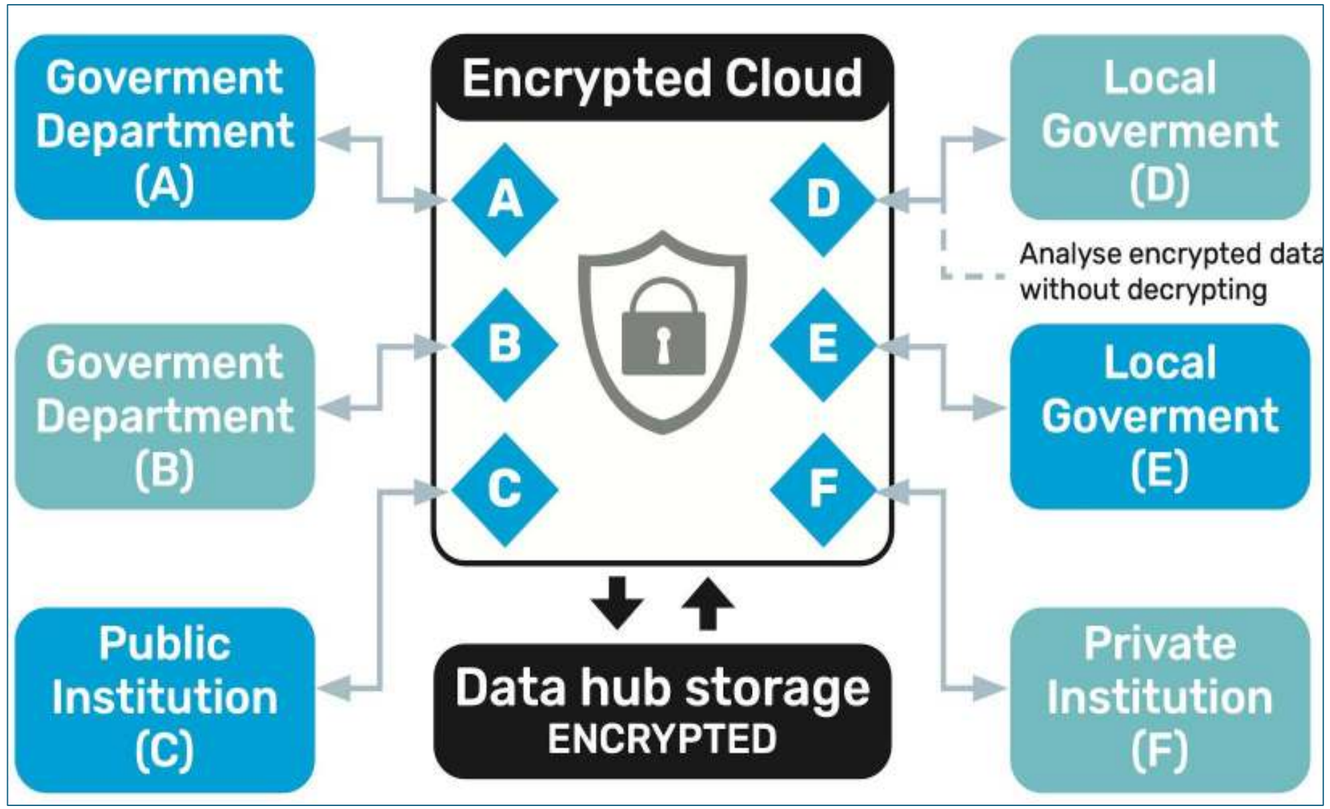
Encrypted Object Detection Model Architecture



Encrypted Object Detection Inference

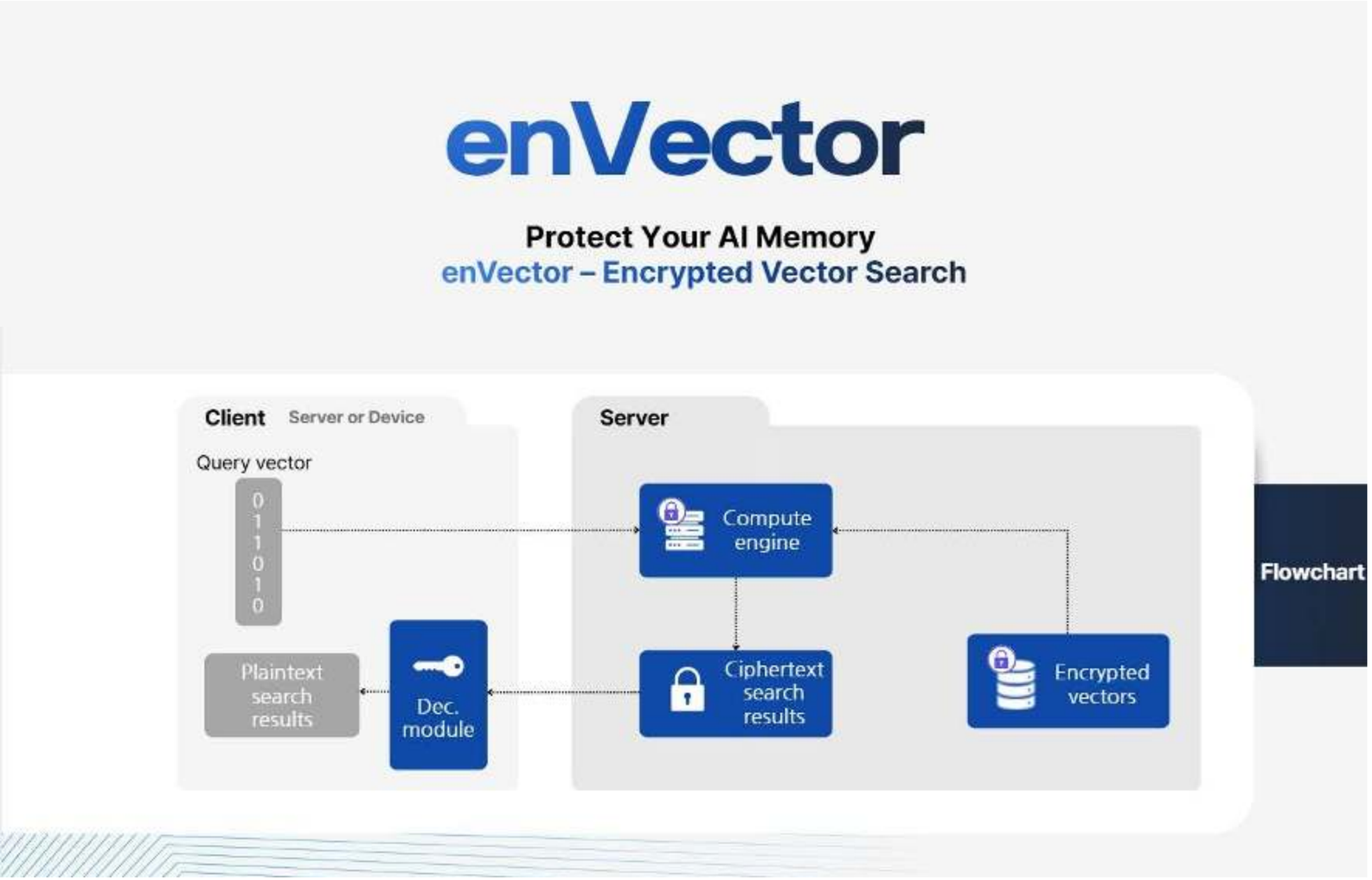
Protects confidential training data—even if the AI model is stolen

[5] Encrypted Data Analytics and Statistics



Collaboration model to create new insights from cross-institutional confidential data

The engine behind the real-world applications: CryptoLab's enVector



- **Minimize data exposure in AI search:** Protects sensitive vectors for secure AI services.
- **Deployment-ready:** No complex security infrastructure required.
- **Search without decryption:** 4.5th gen CKKS homomorphic encryption ensures end-to-end data security.
- **Easy deployment:** Docker image (On-prem & Cloud ready), SaaS (enVector Cloud)

For trial, visit:
envector.io

Part 3: Latest Developments in Encrypted LLMs

Encrypted LLM Implementation

Integrated Implementation

- End-to-end encrypted LLM, optimized via integration of HEaaN2 and 4.5th-gen CKKS (matrix multiplication)

Key Challenges

- Prior FHE LLM results limited to small models (e.g., BERT) or short inputs (8–128 tokens)
- Outliers in large models leading to reduced HE accuracy/efficiency
- Bootstrapping overhead as a major cost driver in end-to-end FHE inference

Encrypted LLM – Key Contributions

01

Matrix Multiplication Optimization

- Appropriate use of multiple matrix-multiplication algorithms
- Design and integration of a new PCMM algorithm for PC-Attention

02

Outlier Mitigation

- Stabilizing activation distributions via Token Prepending [XTC+24]
- Dispersing outlier energy via Orthogonal Rotation [AMC+24]
- 99.7% reduction in the range of the nonlinear function (RMSNorm)

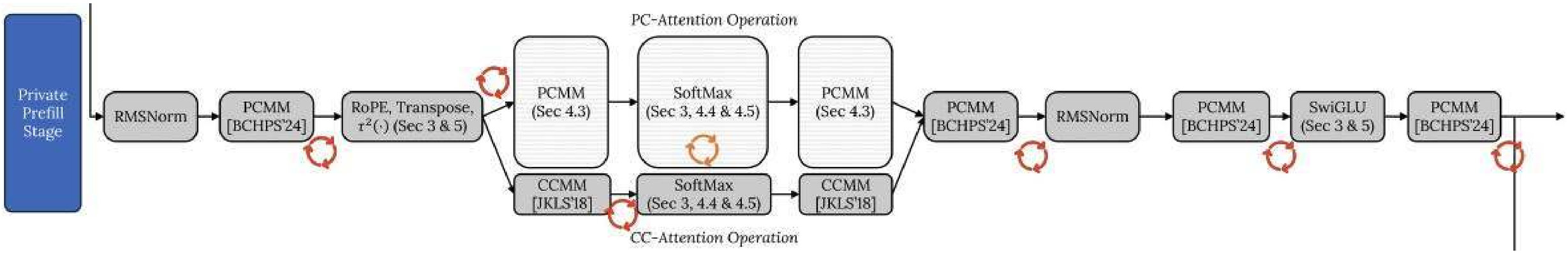
03

Two-Stage Prefill

- Public/Private prefill separation
- Only 128 of 4,096 tokens encrypted
- Reduced ciphertext–ciphertext compute

Matrix Multiplication Optimization

Appropriate use of multiple matrix-multiplication algorithms



- [BCHPS'24] Plaintext-Ciphertext Matrix Multiplication and FHE Bootstrapping: Fast and Fused. CRYPTO 2024. (ePrint 2024/1284)
- [JKLS'18] Secure Outsourced Matrix Computation and Application to Neural Networks. CCS 2018. (ePrint 2018/1041)
- [PPP26] Scaling up Privacy-Preserving ML: A CKKS Implementation of Llama-2-7B (arXiv:2601.18511)

Outlier Mitigation

Input range control with Token Prepending & Orthogonal Rotation

01

Token Prepending

- Prefix token insertion
- Distribution stabilization without retraining
- Prefix derived from public data — no private data required

02

Orthogonal Rotation

- Orthogonal transform to distribute outlier energy across all dimensions
- Fused into the weight matrices with no additional runtime cost
- Preservation of original model outputs

Outlier Reduction Results

	Before	After	Reduction Rate
RMSNorm	2,243.97	7.65	**−99.7%**
SiLU	23.00	10.82	**−53%**
Softmax	39.24	32.78	**−16%**

99.7%
Outlier Reduction

Benchmark Results

16s
Prefill Time

1s
Decode Time (per token)

8×
RTX-5090 GPU

8.4×
Speedup vs. SOTA

	GPU	Encrypted Tokens	Time to First Token
Ours[1 & 2] HEAAN CRYPTO LAB	8 x RTX5090	128	16s (34s)
CERIUM[3] NVIDIA	8 x B200	128	134s

- **8.4× faster inference than prior SOTA**
- **B100-class baseline outperformance on consumer-grade GPUs**

[1] Scaling up Privacy-Preserving ML: A CKKS Implementation of Llama-2-7B (arXiv:2601.18511)
[2] Recent Update with 8 * RTX-5090
[3] Jayashankar et al. “A Scalable Multi-GPU Framework for Encrypted Large-Model Inference”, arXiv:2512.11269

Encrypted LLM will unlock previously impossible use cases

Example use cases of encrypted LLM

Private AI Counseling

- Safely disclose deeply personal issues
- Plaintext-inaccessible inputs (by design)
- Hyper-personalized, privacy-first coaching

Secure Enterprise Intelligence

- Confidential analysis of M&A, trade secrets, proprietary code
- Reduced leakage risk to public cloud / hosted LLM providers (by design)

Confidential Collaborative R&D

- Joint analytics across partners (e.g., pharma / hospitals)
- Collective intelligence without exposing proprietary datasets

Encrypted LLMs will open a new frontier in AI

Upcoming webinar on encrypted LLM

GLOBAL WEBINAR · APRIL 2026

Unlocking the Future of AI Security: Encrypted LLM

An exclusive global webinar hosted by **Prof. Jung Hee Cheon (CEO, CryptoLab)** — exploring how Homomorphic Encryption is redefining privacy in AI.

Following our highly successful **"Homomorphic Encryption and Private AI"** workshop at **Seoul National University**, we are taking the conversation global. Join us this April for an in-depth exploration of **Encrypted Large Language Models (LLMs)** leveraging cutting-edge Homomorphic Encryption.

WHAT YOU WILL LEARN

- The current landscape and future of Private AI
- Practical applications of Homomorphic Encryption in LLMs
- How to ensure absolute data privacy without compromising AI performance

Registration:

Conclusions

1. FHE has reached the state of **real-time processing.**
2. **Real-time, real-world applications** of FHE are spreading.
3. **Real-time Encrypted LLM** is now a reality

Real-Time FHE AI Benchmarks (2 x RTX 5090 GPU)

Category	Model	Architecture	Latency
Face Recognition	ArcFace / AdaFace	IResNet-50, IResNet-100	**2.05s, 3.97s**
Text Embedding	MiniLM	all-MiniLM-L6-v2	**1.45s**
	SBERT	BERT-base	**3.13s**

Thank you.

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Try HEaaN2 Library:

Try enVector:

Join the upcoming webinar on eLLM: